

Bugs found: 4

1. `NullPointerException`: variable 'b' is null when `b.equals(null)` is called.
2. `ArrayIndexOutOfBoundsException`: `arr` is empty, so `arr[0]` is an invalid access.
3. Integer overflow: `Integer.MAX_VALUE * 2` overflows `int` range, producing an incorrect result.
4. Incomplete `equals()` contract: `hashCode()` is overridden in `Bad` but `equals()` is not, violating the `Object` contract.